



3905 - Data-Driven Wisp Templates and Flat Fields for NIRCcam

Cycle: 2, Proposal Category: AR

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Dr. Benjamin D. Johnson (PI)	Harvard University
Dr. Sandro Tacchella (CoI) (ESA Member)	University of Cambridge
Prof. Brant Robertson (CoI)	University of California - Santa Cruz
Dr. Marcia J. Rieke (CoI)	University of Arizona
Dr. Daniel J. Eisenstein (CoI) (CoPI) (US Admin CoI)	Harvard University
Dr. Christopher Nicholas Andrew Willmer (CoI)	University of Arizona
Dr. Chris J. Willott (CoI) (CSA Member)	NRC Herzberg Institute of Astrophysics
Dr. Karl Misselt (CoI)	University of Arizona
Dr. Eiichi Egami (CoI)	University of Arizona
Everett Schlawin (CoI)	University of Arizona

OBSERVATIONS

ABSTRACT

Extracting the most information from JWST NIRCcam observations requires addressing several known issues in the NIRCcam images such as wisps. We propose to use extensive NIRCcam imaging that has been taken during Cycle 1 to construct a large dataset of pixels that have seen only astrophysical and telescope backgrounds. We will use these pixels to extract, via matrix decomposition, a number of NIRCcam background 'templates' that describe the morphology of the wisp straylight and its variation. We will also use the dataset to construct high signal-to-noise 'sky flats', both through robust stacking and from components of the matrix decomposition which naturally account for varying backgrounds. We will look for indications of any other effects that may have stable spatial distributions but vary in amplitude. These analyses will require extensive masking of flux from stars and galaxies. We will release the wisp templates and flat fields to the community along with tools for their use.

OBSERVING DESCRIPTION

This proposal will produce a set of multiple wisp templates for every NIRCcam short-wavelength detector that can be fit to and subtracted from images. The multiple templates will account for any variations in the wisp morphology. They will be derived from aggressively source masked pixels in existing imaging data using matrix decomposition techniques. We will use the same set of source masked pixels to construct 'sky' flats. All templates and tools for fitting them to images will be made available to the community.